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RTable 1. **Quantitative ablation of BNoise and ITB on the IIOF dataset.** We compare restoring VMDiff's BNoise initialization (*w/ BNoise*) with the proposed Image–Text Blending (*w/ Our ITB*). The results show that ITB consistently improves SS and all VLM-based metrics, supporting that the gain comes from the proposed image–text coupling rather than from BNoise initialization.

Methods	**SS** ↑	**CLIP-FlanT5** ↑			**Qwen2.5-VL** ↑			**LLaVA-1.5** ↑		
		P_g	P_{single}	P_{Fuse}	P_g	P_{single}	P_{Fuse}	P_g	P_{single}	P_{Fuse}
w/ BNoise	1.588	0.606	0.423	0.506	0.507	0.186	0.370	0.484	0.331	0.295
w/ Our ITB	**1.736**	**0.647**	**0.466**	**0.562**	**0.618**	**0.252**	**0.468**	**0.520**	**0.359**	**0.332**

RTable 2. **Ablation on early-readout timesteps.** Results on the *strawberry* subset, where *strawberry* is fused with the other objects in the dataset. Runtime is reported in seconds. We mark the selected timestep as $776^{\dagger}$. The results support the use of early readout as an efficient search proxy: compared with near-complete denoising ($t = 089$), $t = 776$ reduces runtime from 71.29s to 33.42s while maintaining a competitive fusion score (SS 2.2431 vs. 2.3110). More generally, image quality improves rapidly in the early-to-middle stage and then saturates, whereas runtime continues to increase substantially.

Timestep	**Time(s)** ↓	**SS** ↑	**CLIP-FlanT5** ↑			**Qwen2.5-VL** ↑			**LLaVA-1.5** ↑		
			P_g	P_{single}	P_{Fuse}	P_g	P_{single}	P_{Fuse}	P_g	P_{single}	P_{Fuse}
999	**9.75**	1.5418	0.6439	0.4416	0.5067	0.5925	0.2607	0.4357	0.5633	0.3341	0.3313
971	13.66	1.7090	0.6377	0.4854	0.5359	0.6361	0.3451	0.4906	0.5603	0.3773	0.3407
943	16.60	1.7534	0.6623	0.5162	0.5742	0.6483	0.3895	0.5467	0.5724	0.4017	0.3650
913	20.23	2.0037	0.6890	0.5782	0.6369	0.6981	0.5212	0.7057	0.5987	0.4617	0.4271
881	23.51	2.0270	0.6825	0.5814	0.6466	0.7374	0.5825	0.7691	0.5938	0.4559	0.4338
848	26.62	2.1644	0.6991	0.6021	0.6628	0.7914	0.6455	0.8245	0.6096	0.4955	0.4579
813	29.97	2.2041	0.6983	0.5927	0.6603	0.7916	0.6313	0.8209	0.6030	0.4851	0.4533
$776^{\dagger}$	33.42	2.2431	0.6958	0.5967	0.6577	0.7967	0.6341	0.8219	0.6011	0.4916	0.4634
736	36.09	2.2193	0.6828	0.5818	0.6439	0.8027	0.5914	0.8031	0.5969	0.4826	0.4512
695	39.21	2.2234	0.7015	0.6052	0.6763	0.7849	0.6131	0.8350	0.6067	0.4788	0.4635
651	43.21	2.2618	0.6972	**0.6139**	0.6754	0.7924	0.6293	0.8317	0.6076	**0.4984**	**0.4674**
604	46.18	2.2705	**0.7051**	0.6062	0.6764	0.7992	0.6084	0.8470	0.6060	0.4825	0.4654
554	50.71	2.2660	0.7021	0.6039	0.6690	0.8054	0.6155	0.8337	0.6050	0.4723	0.4595
501	53.12	2.2471	0.6987	0.5935	0.6641	0.8045	0.6126	0.8330	0.5987	0.4663	0.4445
444	56.62	2.2633	0.7025	0.6003	0.6623	0.8106	0.6187	0.8470	0.6092	0.4695	0.4586
383	59.60	2.2606	0.6972	0.5900	0.6603	**0.8168**	0.6087	0.8273	0.6035	0.4733	0.4523
318	63.17	2.2389	0.6933	0.5904	0.6632	0.8122	0.6018	0.8220	0.6033	0.4749	0.4505
247	67.58	2.2954	0.6933	0.6050	0.6786	0.8127	0.6349	0.8424	**0.6106**	0.4864	0.4648
171	69.29	2.2845	0.7023	0.6094	**0.6804**	0.8162	0.6281	0.8501	0.6088	0.4866	0.4619
089	71.29	**2.3110**	0.6961	0.6119	0.6735	0.8146	**0.6457**	**0.8553**	0.6065	0.4966	0.4635

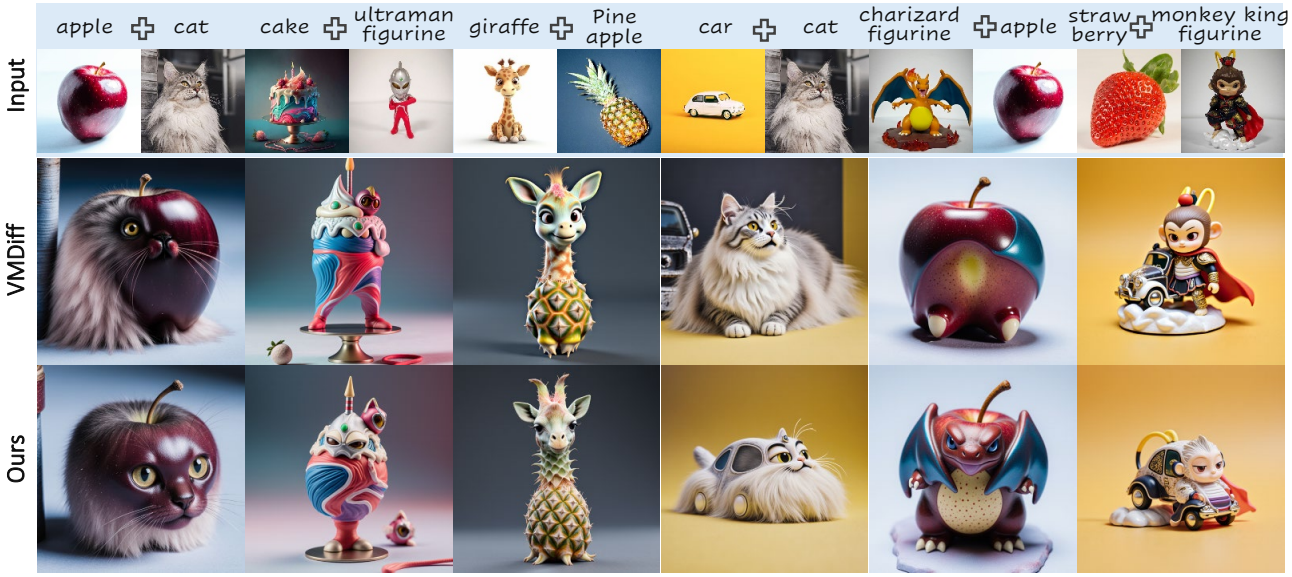
RTable 3. **Direct validation of the ERS ranking assumption across 10 object categories.** To directly test whether the early readout at $t^{\dagger}$=776 preserves the ranking of candidate mixing factors, we evaluate the Spearman rank correlation between the early-readout scores and the fully denoised scores over a fixed grid of $\alpha \in \{0.40, 0.42, \ldots, 0.60\}$ for each image pair. **In total, we evaluate 390 image pairs from 10 object categories.** Since each pair is measured over **11** candidate α values, this gives **4,290** score evaluations in total. For each pair, we compute the rank correlation across all α values, and then average these pair-level correlations within each object category. **The average mean Spearman correlation ρ_s across these 10 categories is 0.8480**, showing that the early readout provides a reliable ranking proxy for selecting α.

Category	**panda**	**astronaut figurine**	**watermelon**	**teapot**	**panda figurine**	**orange**	**monkey king figurine**	**horse**	**corgi**	**doll figurine**	**Average**
Mean ρ_s	0.8825	0.8781	0.8744	0.8662	0.8652	0.8531	0.8266	0.8202	0.8109	0.8026	**0.8480**

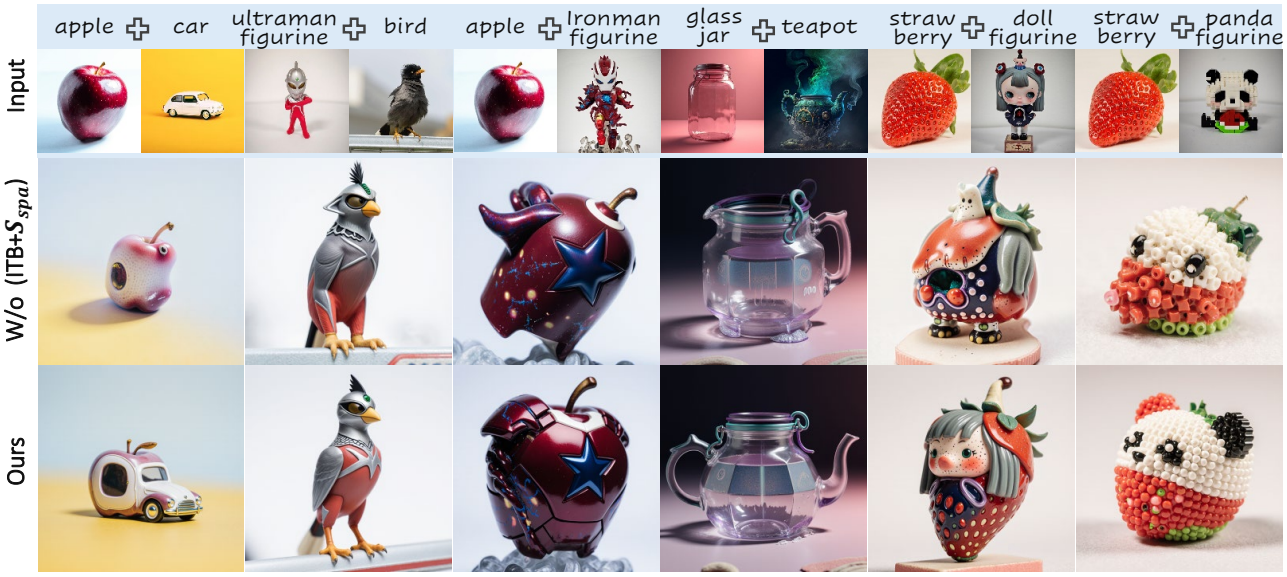
RTable 4. **Multi-seed robustness of ITMDiff on the evaluation subset.** We report SS and VQA-style metrics under four random seeds $\{42, 99, 66, 1\}$. The results are overall stable across seeds, indicating that the fixed-seed setting used in the main comparisons is not a specially favorable case.

Seed	**SS** ↑	**CLIP-FlanT5** ↑			**Qwen2.5-VL** ↑			**LLaVA-1.5** ↑		
		P_g	P_{single}	P_{Fuse}	P_g	P_{single}	P_{Fuse}	P_g	P_{single}	P_{Fuse}
Seed 42	2.069	0.671	0.544	**0.666**	**0.707**	**0.447**	0.664	0.586	**0.445**	0.427
Seed 99	2.060	0.671	0.534	0.648	0.701	0.427	0.651	0.588	0.434	0.420
Seed 66	**2.072**	**0.674**	**0.547**	0.661	0.703	0.422	**0.676**	0.583	0.441	**0.428**
Seed 1	2.036	0.670	0.529	0.647	0.681	0.410	0.634	**0.588**	0.436	0.418

RFigure 1. **Preliminary three-image fusion results.** We extend ITMDiff to three inputs using a simple sequential strategy: two images are first fused into an intermediate hybrid, which is then fused with a third image. The results suggest that the framework can be extended beyond pairwise fusion, while also showing more noticeable information loss and imbalance as the number of inputs increases.



RFigure 2. **Additional qualitative comparison with VMDiff on more object pairs.** ITMDiff more consistently produces a *single coherent object* that integrates shape, structure, and material cues from both inputs. In contrast, VMDiff often relies more on local texture transfer, partial part attachment, or residual co-location, which leads to weaker holistic fusion.



RFigure 3. **Qualitative results without explicit object category names.** We remove the name-dependent components, i.e., ITB token re-scaling and $S_{\mathbf{spa}}$, and keep only image-based fusion. The model can still generate plausible fused objects, but the results more often lose distinctive details or become biased toward one source, showing that category names serve as useful semantic anchors for better detail preservation and balanced fusion.